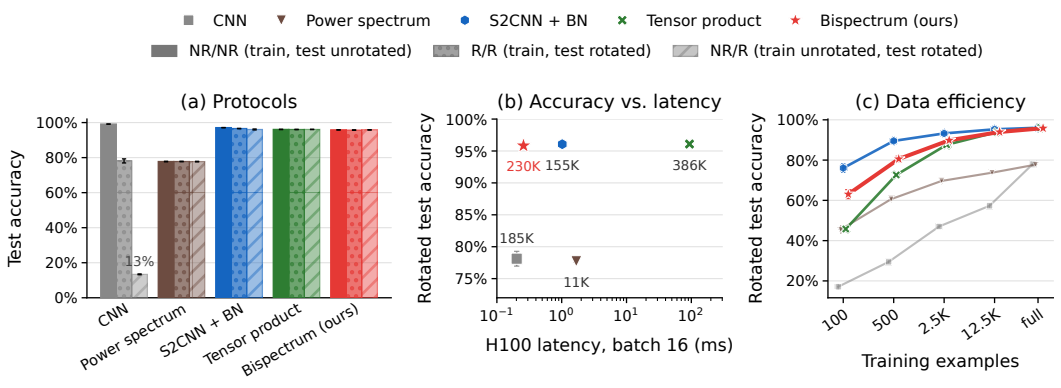

CNN
Power spectrum
S2CNN + BN
Tensor product
Bispectrum (ours)
NR/NR (train, test unrotated)
R/R (train, test rotated)
NR/R (train unrotated, test rotated)
(a) Protocols
Test accuracy
100%
80%
60%
40%
20%
0%
13%
CNN
Power spectrum
S2CNN + BN
Tensor product
Bispectrum (ours)
(b) Accuracy vs. latency
Rotated test accuracy
100%
95%
90%
85%
80%
75%
230K
155K
386K
185K
11K
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
H100 latency, batch 16 (ms)
(c) Data efficiency
Rotated test accuracy
100%
80%
60%
40%
20%
100
500
2.5K
12.5K
full
Training examples